

Computer Science — One-Line Answers

Total Questions: 521

Q1. What does an operating system do?

→ It manages hardware resources and provides services for computer programs.

Q2. What does FAT mean?

→ FAT stands for File Allocation Table, a file system used to track disk clusters.

Q3. Which one of the following is not true?

→ That the OS does not manage memory — the OS does manage memory.

Q4. What is Interprocess communication?

→ IPC is a mechanism that allows processes to communicate and synchronize with each other.

Q5. Messages sent by a process

→ Messages sent by a process are used in message-passing IPC to exchange data between processes.

Q6. In shared Memory

→ In shared memory IPC, processes share a common memory region for direct data access.

Q7. Bounded capacity and Unbounded capacity queues are referred to as

→ They are referred to as automatic buffering.

Q8. When several processes access the same data concurrently and the outcome depends on the order is called

→ This is called a Race Condition.

Q9. If a process is executing in its critical section, then no other processes can be executing in their critical section. What is this condition called?

→ This condition is called Mutual Exclusion.

Q10. Mutual exclusion can be provided by the

→ Mutual exclusion can be provided by semaphores, mutex locks, or monitors.

Q11. Process synchronization can be done on

→ Process synchronization can be done on shared memory or message-passing systems.

Q12. Peterson's solution is restricted to ____ processes

→ Peterson's solution is restricted to 2 processes.

Q13. We use the mutex lock to protect critical regions; the term mutex is short for

→ Mutex is short for Mutual Exclusion.

Q14. A system is in the safe state if

→ A system is in the safe state if there exists a safe sequence of all processes.

Q15. A classic _____ solution to the critical-section problem known as Peterson's solution

→ A classic software solution to the critical-section problem.

Q16. Which one of the following is the deadlock avoidance algorithm?

→ Banker's Algorithm is the deadlock avoidance algorithm.

Q17. What is the drawback of banker's algorithm?

→ It requires advance knowledge of maximum resource needs, which is impractical.

Q18. A problem encountered in multitasking when a process is perpetually denied necessary resources is called

→ This is called Starvation.

Q19. Which one of the following is a visual (mathematical) way to determine deadlock occurrence?

→ Resource Allocation Graph is used to determine deadlock occurrence.

Q20. If the resources are always preempted from the same process _____ can occur.

→ Starvation can occur.

Q21. In segmentation, each address is specified by

→ Each address is specified by a segment number and an offset.

Q22. If there are 32 segments, each of size 1Kb, then the logical address should have

→ The logical address should have 5 bits for segment number and 10 bits for offset (15 bits total).

Q23. A multilevel page table is preferred over a single level page table because

→ It saves memory by not allocating page table space for unused address regions.

Q24. The offset 'd' of the logical address must be

→ The offset 'd' must be less than the segment length.

Q25. In paging the user provides only _____ which is partitioned by the hardware into _____ and _____

→ The user provides only the logical address, partitioned into page number and page offset.

Q26. Consider a computer with 8 Mbytes of main memory and 128K cache. Block size 4K. Direct mapping. How many main memory blocks can map onto a given cache block?

→ 32 different main memory blocks can map onto a given cache block.

Q27. Which of the architecture is power efficient?

→ RISC (Reduced Instruction Set Computer) architecture is more power efficient.

Q28. What is computer architecture?

→ Computer architecture is the design and organization of a computer's functional components.

Q29. Which of the following is a type of computer architecture?

→ Von Neumann architecture is a type of computer architecture.

Q30. What does CSA stand for?

→ CSA stands for Carry Save Adder.

Q31. Floating point representation is used to store

→ Floating point representation is used to store real numbers with fractional parts.

Q32. Assembly language

→ Assembly language is a low-level programming language using symbolic mnemonics for machine instructions.

Q33. In computers, subtraction is generally carried out by

→ Subtraction is carried out by adding the two's complement of the subtrahend.

Q34. To reduce the memory access time we generally make use of

→ Cache memory is used to reduce memory access time.

Q35. The main virtue for using single Bus structure is

→ The main virtue is simplicity and low cost.

Q36. To extend the connectivity of the processor bus we use

→ We use a bus extender or bridge to extend processor bus connectivity.

Q37. The register Connected to the Processor bus is a single-way transfer capable?

→ No, it is bidirectional (capable of both read and write transfers).

Q38. In multiple Bus organisation, the registers are collectively placed and referred as

→ They are referred to as a Register File.

Q39. _____ are used to overcome the difference in data transfer speeds of various devices.

→ Buffers are used to overcome differences in data transfer speeds.

Q40. IBM developed a bus standard for their line of computers 'PC AT' called

→ IBM developed the ISA (Industry Standard Architecture) bus standard.

Q41. The ALU gives the output and the output is stored in the

→ The output is stored in the Accumulator register.

Q42. The process of division on memory spaces is called

→ It is called Memory Segmentation.

Q43. The bitwise complement of 0 is

→ The bitwise complement of 0 is 1.

Q44. Which of the following is used for binary multiplication?

→ Booth's Algorithm is used for binary multiplication.

Q45. If Booth's Multiplication is performed on numbers 22×3 , then what is 3 referred to as

→ 3 is referred to as the Multiplier.

Q46. What is the default value of accumulator in booth's multiplication of two 4-bit binary numbers?

→ The default value of the accumulator is 0000 (all zeros).

Q47. Which one of the following is a multiplication algorithm?

→ Booth's Algorithm is a multiplication algorithm.

Q48. Booth algorithm gives procedure for multiplying binary integers in

→ Booth's algorithm multiplies binary integers in signed-2's complement representation.

Q49. In a half-adder, CARRY is obtained by using

→ CARRY is obtained by using an AND gate.

Q50. In multiplication algorithm, after initialization the _____ bit of the multiplier in Q_n is tested.

→ The LSB (Least Significant Bit) of the multiplier is tested.

Q51. Which shift is used in multiplication algorithm?

→ Arithmetic right shift is used in the multiplication algorithm.

Q52. In multiplication algorithm, shifting operation is done on?

→ Shifting is done on the combined AC (accumulator) and QR (multiplier) register pair.

Q53. The name hardwired came because the sequence of operations carried out is determined by the wiring

→ True — the control sequence is fixed by physical circuit connections.

Q54. The disadvantage/s of the hardwired approach is

→ It is inflexible; modifying the control logic requires redesigning the hardware.

Q55. The advantage of I/O mapped devices to memory mapped is

→ I/O mapped devices use a separate address space, freeing memory addresses.

Q56. To overcome the lag in the operating speeds of the I/O device and the processor we use

→ We use Buffering or DMA (Direct Memory Access).

Q57. The method of accessing the I/O devices by repeatedly checking the status flags is

→ This method is called Polling (Programmed I/O).

Q58. The method of synchronising the processor with the I/O device in which the device sends a signal when it is ready is?

→ This method is called Interrupt-driven I/O.

Q59. Whenever the data is found in the cache memory it is called as

→ It is called a Cache Hit.

Q60. When the data at a location in cache is different from main memory, the cache is called

→ The cache is called Dirty or Inconsistent.

Q61. In _____ mapping, the data can be mapped anywhere in the Cache Memory.

→ In Fully Associative mapping, data can be mapped anywhere in cache.

Q62. The effectiveness of the cache memory is based on the property of

→ It is based on the principle of Locality of Reference.

Q63. The correspondence between main memory blocks and cache is given by

→ It is given by the Mapping Function.

Q64. Who created the first DBMS?

→ Charles Bachman created the first DBMS (IDS) in the early 1960s.

Q65. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

→ It is known as an Entity Set.

Q66. What is a relation in RDBMS?

→ A relation is a two-dimensional table consisting of rows (tuples) and columns (attributes).

Q67. Which of the following is the full form of RDBMS?

→ RDBMS stands for Relational Database Management System.

Q68. What is a database?

→ A database is an organized collection of structured data stored and accessed electronically.

Q69. Which type of data can be stored in the database?

→ Structured, semi-structured, and unstructured data can be stored in a database.

Q70. If a set is a collection of values given by the select clause, The _____ connective tests for set membership

→ The IN connective tests for set membership.

Q71. State true or false: Nested Subqueries cannot be used for comparing two different sets

→ False — nested subqueries can be used for comparing two different sets.

Q72. What is a correlated sub-query?

→ B) A sub-query that uses the correlation name of an outer query.

Q73. The _____ construct returns true if the argument in the sub-query is void of duplicates

→ The UNIQUE construct returns true if the subquery has no duplicates.

Q74. SQL subqueries that return only one tuple with a single attribute are called

→ They are called Scalar Subqueries.

Q75. What is a subquery?

→ A) A subquery is a select-from-where expression nested within another query.

Q76. Which of the following is TRUE regarding normal forms?

→ C) Every relation in BCNF is also in 3NF.

Q77. We can use the following three rules to find logically implied functional dependencies. This collection of rules is called

→ This collection is called Armstrong's Axioms.

Q78. Which of the following is not Armstrong's Axiom?

→ C) Pseudotransitivity rule is not a basic Armstrong's Axiom (it is derived).

Q79. We can use the following three rules to find logically implied functional dependencies. This collection is called

→ Armstrong's Axioms (Reflexivity, Augmentation, Transitivity).

Q80. A functional dependency f on R is _____ by a set F if every instance satisfying F also satisfies f

→ The FD f is logically implied by F.

Q81. If F is a set of functional dependencies, the closure of F is denoted by?

→ The closure of F is denoted by F⁺.

Q82. If a functional dependency is reflexive, B is a subset of A, then

→ Then A → B holds trivially (reflexivity).

Q83. Armstrong axioms are called sound because?

→ They generate only valid (correct) functional dependencies from a given set.

Q84. A management procedure called _____ control is required to control concurrent database operations.

→ Concurrency control is required.

Q85. The concurrent execution of the database in a _____-user system

→ Multi-user system allows concurrent database execution.

Q86. When dealing with database transactions, multiple users accessing the same database causes

→ Concurrent execution of the database occurs.

Q87. During simultaneous execution, operations performed interleaved maintains a database

→ It maintains database consistency (serializability).

Q88. How many concurrency problems are there?

→ There are 3 main concurrency problems: lost update, dirty read, and unrepeatable read.

Q89. Which of the following is a concurrency problem?

→ Dirty Read is a concurrency problem.

Q90. If a transaction has obtained a _____ lock, it can read but cannot write on the item

→ A Shared (Read) lock allows reading but not writing.

Q91. A transaction can proceed only after the concurrency control manager _____ the lock

→ The manager grants the lock to the transaction.

Q92. If a lock can be granted immediately despite another mode, the two modes are said to be

→ The two modes are said to be Compatible.

Q93. The situation where no transaction can proceed with normal execution is known as

→ This is known as Deadlock.

Q94. The protocol indicating when a transaction may lock and unlock data items is called

→ It is called the Locking Protocol.

Q95. If a transaction may release locks but may not obtain any locks, it is in _____ phase

→ It is in the Shrinking Phase.

Q96. What is the time complexity of binary search in the worst-case scenario

→ $O(\log n)$ is the worst-case time complexity of binary search.

Q97. Which data structure is best suited for LIFO behavior?

→ A Stack is best suited for LIFO (Last In, First Out) behavior.

Q98. In a sparse matrix representation, which data structure stores non-zero elements?

→ A Linked List or array of triples is commonly used to store non-zero elements.

Q99. Which sorting algorithm has a worst-case time complexity of $O(n^2)$?

→ Bubble Sort, Insertion Sort, and Selection Sort all have $O(n^2)$ worst-case complexity.

Q100. In a queue, which operation inserts an element at the rear end?

→ The Enqueue operation inserts an element at the rear end.

Q101. Which of the following is true about depth-first search (DFS)?

→ DFS uses a stack (or recursion) and may not find the shortest path.

Q102. Which is a disadvantage of a singly linked list compared to a doubly linked list

→ A singly linked list cannot be traversed in reverse direction.

Q103. Which type of linked list allows traversal from the last node to the first node

→ A Doubly Linked List or Circular Doubly Linked List allows reverse traversal.

Q104. Which of the following is a binary tree where each node has at most two children

→ A Binary Tree is a tree where each node has at most two children.

Q105. Which of the following is true about a binary search tree (BST)

→ In a BST, the left subtree contains values less than the root, and right subtree contains greater values.

Q106. Which data structure is used for implementing priority queues

→ A Heap (Min-Heap or Max-Heap) is used for implementing priority queues.

Q107. Which data structure is used for implementing depth-first search (DFS)

→ A Stack is used for implementing DFS.

Q108. Which of the following is NOT an application of graphs in data structures

→ Sorting of elements is NOT a typical application of graphs.

Q109. In which scenario would a graph data structure be most suitable (1)

→ Modeling a social network where users are nodes and relationships are edges.

Q110. In which scenario would a graph data structure be most suitable (2)

→ Finding the shortest route in a city map/navigation system.

Q111. Which algorithm is commonly used to find the shortest path in a weighted graph

→ Dijkstra's Algorithm is used for shortest path in a weighted graph.

Q112. Which of the following is an example of an unweighted graph

→ A social network (friendship graph) is an example of an unweighted graph.

Q113. Which data structure is typically used to implement graphs efficiently

→ An Adjacency List is typically used to implement graphs efficiently.

Q114. Which algorithm is commonly used to detect cycles in a graph

→ DFS (Depth-First Search) with a visited set is used to detect cycles.

Q115. Which of the following is not a comparison-based sorting algorithm

→ Counting Sort, Radix Sort, and Bucket Sort are not comparison-based.

Q116. Which sorting algorithm has the best average-case time complexity

→ Merge Sort and Quick Sort have the best average-case time complexity of $O(n \log n)$.

Q117. Which sorting algorithm is most suitable for sorting a linked list

→ Merge Sort is most suitable for sorting a linked list.

Q118. Which of the following is NOT a characteristic of a good hash function

→ A good hash function should NOT produce many collisions.

Q119. Which hashing technique is vulnerable to clustering

→ Linear Probing is vulnerable to clustering.

Q120. Which sorting algorithm does not depend on swapping elements directly to sort (1)

→ Merge Sort does not depend on swapping elements directly.

Q121. Which sorting algorithm does not depend on swapping elements directly to sort (2)

→ Merge Sort — it merges sorted sublists without direct swaps.

Q122. Which sorting algorithm is most efficient for small datasets and partially sorted arrays

→ Insertion Sort is most efficient for small and partially sorted arrays.

Q123. Which of the following is NOT an in-place sorting algorithm

→ Merge Sort is NOT an in-place sorting algorithm.

Q124. Which hashing technique uses chaining to resolve collisions

→ Separate Chaining uses linked lists to resolve collisions.

Q125. Which sorting algorithm typically uses the least amount of extra memory space

→ Heap Sort or Quick Sort (in-place) uses the least extra memory.

Q126. The transition function of an NFA is defined as

→ $\delta: Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$ (maps state and input to a set of states).

Q127. The power set construction is used to convert

→ Power set construction converts an NFA to an equivalent DFA.

Q128. The main advantage of NFA over DFA is

→ NFA is more concise and easier to construct for certain languages.

Q129. NFA stands for

→ NFA stands for Nondeterministic Finite Automaton.

Q130. The transition function in a DFA is defined as

→ $\delta: Q \times \Sigma \rightarrow Q$ (maps a state and input symbol to exactly one state).

Q131. What does ϵ symbolize in an NFA with ϵ transitions

→ ϵ symbolizes an empty (null) transition that requires no input symbol.

Q132. What does DFA stand for

→ DFA stands for Deterministic Finite Automaton.

Q133. Which of the following machines can accept only regular languages

→ Finite Automaton (DFA/NFA) accepts only regular languages.

Q134. Regular grammar generates

→ Regular grammar generates regular languages.

Q135. Which of the following is NOT a component of a regular grammar

→ A pushdown stack is NOT a component of a regular grammar.

Q136. Pumping Lemma is used to prove that a language is

→ Pumping Lemma is used to prove that a language is NOT regular.

Q137. Which statement is true regarding the equivalence of DFA and NFA

→ DFA and NFA are equivalent in expressive power — both accept regular languages.

Q138. The Chomsky Normal Form (CNF) simplifies parsing by

→ CNF simplifies parsing by restricting rules to $A \rightarrow BC$ or $A \rightarrow a$ form.

Q139. Ambiguity in context-free grammars can lead to

→ Ambiguity can lead to multiple parse trees for the same string.

Q140. Context-free grammars are used to describe languages recognized by

→ CFGs describe languages recognized by Pushdown Automata (PDA).

Q141. The Myhill-Nerode theorem is based on the idea of distinguishing:

→ It is based on distinguishing strings by their future behavior (equivalence classes).

Q142. In DFA state minimization, two states are considered equivalent if:

→ Two states are equivalent if they are indistinguishable for all input strings.

Q143. Greibach Normal Form (GNF) allows production rules to have

→ GNF allows rules of the form $A \rightarrow a\alpha$ where a is a terminal and α is a string of variables.

Q144. Nondeterministic Pushdown Automata (NDPA) is an extension of:

→ NDPA is an extension of Deterministic Pushdown Automata (DPDA).

Q145. The primary difference between a PDA and a NPDA is

→ NPDA allows multiple possible transitions, while PDA has at most one.

Q146. A NPDA with multiple possible transitions is an example of

→ It is an example of Nondeterminism.

Q147. Pumping Lemma for Context-Free Languages is used to

→ It is used to prove that a language is NOT context-free.

Q148. Closure properties of Context-Free Languages include

→ CFLs are closed under union, concatenation, and Kleene star.

Q149. Context-Sensitive Grammar (CSG) is a type of grammar that generates

→ CSG generates context-sensitive languages (Type-1 in Chomsky hierarchy).

Q150. Linear Bounded Automata (LBA) are a restricted form of:

→ LBA are a restricted form of Turing Machines.

Q151. The tape of a Linear Bounded Automaton (LBA) is:

→ The tape is bounded to the length of the input string.

Q152. Recursive languages are closed under:

→ Recursive languages are closed under complement, union, and intersection.

Q153. The Chomsky hierarchy provides a framework for classifying:

→ It classifies formal languages and their corresponding grammars and automata.

Q154. Turing-recognizable languages are also known as:

→ They are also known as Recursively Enumerable languages.

Q155. Linear Bounded Automata (LBA) have limitations in terms of:

→ LBA are limited by tape length equal to input length.

Q156. A language is recursively enumerable if:

→ A language is recursively enumerable if a Turing Machine accepts it (may not halt on rejection).

Q157. BIOS is used?

→ BIOS is used to initialize hardware and boot the operating system at startup.

Q158. In infix to postfix conversion, which operator has the highest precedence

→ Exponentiation (^) has the highest precedence.

Q159. Which data structure is commonly used to implement a stack

→ An Array or Linked List is commonly used to implement a stack.

Q160. Which data structure is typically used to evaluate postfix expressions

→ A Stack is used to evaluate postfix expressions.

Q161. What does the term 'asymptotic notation' refer to in algorithm analysis?

→ Asymptotic notation describes the limiting behavior (growth rate) of an algorithm's time/space complexity.

Q162. Which notation represents the worst-case time complexity of an algorithm?

→ Big-O notation (O) represents the worst-case time complexity.

Q163. Space complexity includes the memory required for which of the following?

→ Space complexity includes memory for variables, data structures, inputs, and function call stack.

Q164. Which data structure is best suited for expression evaluation?

→ A Stack is best suited for expression evaluation.

Q165. A sparse matrix is one where the number of non-zero elements is:

→ A sparse matrix has very few non-zero elements compared to total elements.

Q166. Which of the following is NOT an advantage of using stacks?

→ Random access to elements is NOT an advantage of stacks.

Q167. Which operation can be efficiently implemented using a queue?

→ Breadth-First Search (BFS) can be efficiently implemented using a queue.

Q168. Which data structure allows insertion and deletion at both ends?

→ A Deque (Double-Ended Queue) allows insertion and deletion at both ends.

Q169. Which of the following is NOT a stack application?

→ CPU scheduling is NOT a typical stack application.

Q170. In a queue, deletion occurs at the:

→ Deletion occurs at the Front of the queue.

Q171. Which statement is true for a singly linked list?

→ Each node has a data field and a pointer to the next node only.

Q172. What is a circular linked list?

→ A circular linked list is one where the last node points back to the first node.

Q173. Which traversal visits nodes: left subtree → root → right subtree?

→ In-order traversal visits in the order: left → root → right.

Q174. In a BST, the left child of a node contains:

→ The left child contains a value less than the parent node.

Q175. What is the maximum number of children a binary tree node can have?

→ A binary tree node can have at most 2 children.

Q176. Which of the following is an application of graphs?

→ Finding shortest paths in networks is an application of graphs.

Q177. Dijkstra's algorithm is used for:

→ Dijkstra's algorithm is used for finding the shortest path in a weighted graph.

Q178. Which graph traversal method uses a queue?

→ Breadth-First Search (BFS) uses a queue.

Q179. Which data structure is best suited for implementing Dijkstra's algorithm efficiently?

→ A Priority Queue (Min-Heap) is best suited for Dijkstra's algorithm.

Q180. Which algorithm finds the minimum spanning tree?

→ Prim's and Kruskal's algorithms find the minimum spanning tree.

Q181. Which sorting algorithm has worst-case time complexity of $O(n^2)$?

→ Bubble Sort, Selection Sort, and Insertion Sort have $O(n^2)$ worst-case.

Q182. Which sorting algorithm is best suited for nearly sorted data?

→ Insertion Sort is best suited for nearly sorted data.

Q183. Which hashing technique minimizes collisions by placing elements in next available slot?

→ Linear Probing minimizes collisions using next available slot.

Q184. Which collision resolution technique uses linked lists?

→ Separate Chaining uses linked lists for collision resolution.

Q185. Which sorting technique is not based on comparisons?

→ Counting Sort, Radix Sort, and Bucket Sort are not comparison-based.

Q186. Which of the following is NOT a type of system call?

→ Compiler call is NOT a type of system call.

Q187. Which OS structure follows a strict hierarchical approach?

→ Layered OS structure follows a strict hierarchical approach.

Q188. What is the primary advantage of using a microkernel architecture?

→ It provides better modularity, security, and fault isolation.

Q189. Which component of the OS handles communication between user applications and hardware?

→ The System Call Interface handles communication between applications and hardware.

Q190. Which system call is used to create a new process in Unix-based systems?

→ The `fork()` system call creates a new process in Unix.

Q191. Which scheduling algorithm may cause starvation?

→ Priority Scheduling may cause starvation of low-priority processes.

Q192. The main reason for race conditions is:

→ Race conditions occur due to uncontrolled concurrent access to shared resources.

Q193. The critical section problem is a concern in:

→ It is a concern in concurrent/multithreaded programming environments.

Q194. Which mechanism is used to prevent race conditions?

→ Mutex locks and semaphores are used to prevent race conditions.

Q195. Which of the following is NOT a condition for deadlock?

→ Preemption is NOT a condition for deadlock; no preemption IS a condition.

Q196. Which synchronization mechanism allows multiple readers but only one writer?

→ Read-Write Lock allows multiple readers but only one writer.

Q197. Which of the following is NOT a classical synchronization problem?

→ Sleeping Barber's Problem is NOT a classical synchronization problem (Dining Philosophers, Producer-Consumer, and Readers-Writers are classical).

Q198. Which synchronization primitive allows busy-waiting?

→ Spinlock allows busy-waiting.

Q199. Peterson's solution is used for:

→ Peterson's solution is used for mutual exclusion between two processes.

Q200. A semaphore value of 0 means:

→ A semaphore value of 0 means no resources are available; processes must wait.

Q201. Which algorithm is used for deadlock avoidance?

→ Banker's Algorithm is used for deadlock avoidance.

Q202. Which of the following is NOT a memory management technique?

→ Defragmentation is not a primary memory management technique.

Q203. Fixed partitioning leads to:

→ Fixed partitioning leads to internal fragmentation.

Q204. Which memory allocation method gives the first available block big enough?

→ First Fit allocation gives the first block that is big enough.

Q205. What is the primary disadvantage of paging?

→ The primary disadvantage is internal fragmentation.

Q206. A language for which no DFA exists is a

→ A non-regular language is one for which no DFA exists.

Q207. A DFA cannot be represented in the following format

→ A DFA cannot be represented as a non-deterministic transition table.

Q208. Consider a DFA with 5 states over {0,1}. Maximum number of distinct strings of length 3 it can accept?

→ Maximum 8 distinct strings of length 3 ($2^3 = 8$) can be accepted.

Q209. How many minimum states are required to design a DFA that accepts all binary strings divisible by 3?

→ Minimum 3 states are required.

Q210. For an NFA with ϵ transitions, which of the following is TRUE?

→ ϵ -closure of a state includes all states reachable via ϵ transitions.

Q211. Which of the following languages is accepted by some DFA?

→ The set of all binary strings with even number of 1s is accepted by a DFA.

Q212. Which of the following languages is NOT regular?

→ The language $\{a^n b^n \mid n \geq 0\}$ is NOT regular.

Q213. Which statement about the Pumping Lemma is correct?

→ If a language fails the Pumping Lemma, it is not regular.

Q214. Which operation is NOT closed for regular languages?

→ Regular languages are closed under all standard operations; complementation, union, intersection, and concatenation are all closed.

Q215. Which of the following is NOT a functional unit of a computer?

→ A compiler is NOT a functional unit of a computer.

Q216. During the fetch phase of the instruction cycle, the CPU fetches:

→ The CPU fetches the instruction from memory using the program counter.

Q217. Which bus is bidirectional?

→ The Data Bus is bidirectional.

Q218. The process of executing instructions one after another is called:

→ It is called Sequential Execution.

Q219. In immediate addressing mode, the operand is:

→ The operand is part of the instruction itself.

Q220. The instruction cycle consists of:

→ The instruction cycle consists of Fetch, Decode, Execute, and Write-back stages.

Q221. The register that holds the address of the next instruction to be executed is:

→ The Program Counter (PC) holds the address of the next instruction.

Q222. What is the primary disadvantage of a single bus architecture?

→ A single bus creates a bottleneck, limiting performance and bandwidth.

Q223. The multiple-bus architecture improves performance by:

→ It allows multiple simultaneous data transfers between components.

Q224. In a processor, which unit is responsible for controlling data flow?

→ The Control Unit is responsible for controlling data flow.

Q225. Which type of processor organization allows multiple instructions to execute in parallel?

→ Pipelined or Superscalar processor organization allows parallel execution.

Q226. In a multiple-bus system, buses primarily help in:

→ Buses help in transferring data between registers, memory, and I/O simultaneously.

Q227. What is the role of an accumulator in processor design?

→ The accumulator stores intermediate results of ALU operations.

Q228. In a multiple-bus system, how does the system handle simultaneous data transfers?

→ Each bus handles a different transfer simultaneously, increasing throughput.

Q229. Which operation does NOT typically require an ALU computation?

→ Memory address generation via direct addressing does not require ALU computation.

Q230. In a single-bus architecture, how does the CPU ensure correct data transfer?

→ It uses bus control signals and arbitration to sequence transfers.

Q231. What is the purpose of a multiplexer in a logic circuit?

→ A multiplexer selects one of many inputs and routes it to a single output.

Q232. Which of the following is a universal logic gate?

→ NAND and NOR are universal logic gates.

Q233. Which is NOT a typical operation performed by an ALU?

→ Memory management is NOT a typical ALU operation.

Q234. Which is a critical component of an ALU?

→ The adder circuit is a critical component of an ALU.

Q235. What is the significance of the zero flag in an ALU?

→ The zero flag indicates that the result of an operation is zero.

Q236. Which of the following is a common flag in a status register?

→ Carry, Zero, Sign, and Overflow are common flags in a status register.

Q237. ALU performs $A+B$ where $A=0111$ and $B=1011$. What is the value of the carry flag?

→ $0111 + 1011 = 10010$, so the carry flag is 1.

Q238. What is the output of a 4-bit left shifter if input is 0011 and shifted left by 2 bits?

→ 0011 shifted left by 2 bits = 1100.

Q239. An ALU is designed to perform 8 operations. How many select lines are required?

→ 3 select lines are required ($2^3 = 8$).

Q240. What is the output of a 3-input XOR gate if $A=1$, $B=1$, $C=0$?

→ $1 \text{ XOR } 1 \text{ XOR } 0 = 0$.

Q241. What is the primary goal of pipelining in processors?

→ Pipelining aims to increase instruction throughput by overlapping execution stages.

Q242. What is the speedup achieved by an n-stage pipeline over non-pipelined processor?

→ Ideal speedup is n times for an n-stage pipeline.

Q243. Which of the following is a disadvantage of pipelining?

→ Pipeline hazards (data, control, structural) can reduce speedup.

Q244. What is the main cause of pipeline hazards?

→ Dependencies between instructions and resource conflicts cause pipeline hazards.

Q245. What is the purpose of pipeline registers in a pipelined processor?

→ Pipeline registers hold intermediate results between pipeline stages.

Q246. What is the main cause of a control hazard in pipelining?

→ Branch instructions cause control hazards due to uncertain next instruction.

Q247. In a 6-stage pipeline, what is the maximum number of instructions in execution simultaneously?

→ Maximum 6 instructions can be in execution simultaneously.

Q248. What is the main disadvantage of increasing the number of pipeline stages?

→ More stages increase pipeline hazards and the cost of branch mispredictions.

Q249. What is the primary purpose of an I/O interface in a computer system?

→ An I/O interface bridges communication between the CPU and I/O devices.

Q250. Which of the following is a type of interrupt?

→ Hardware interrupt, software interrupt (trap), and maskable/non-maskable interrupts.

Q251. What is the main disadvantage of using interrupts?

→ Interrupts add overhead due to context switching and interrupt handling.

Q252. What is the main advantage of DMA over programmed I/O?

→ DMA transfers data directly between memory and device without CPU involvement.

Q253. What is the main component used to implement hardwired control?

→ Combinational logic circuits (gates and flip-flops) implement hardwired control.

Q254. What is the role of the DMA controller?

→ The DMA controller manages direct data transfer between I/O devices and memory.

Q255. Which is true about hardwired control units?

→ Hardwired control is fast but inflexible; changes require hardware redesign.

Q256. What is the purpose of the interrupt mask register?

→ The interrupt mask register enables or disables specific interrupts.

Q257. Which of the following is a type of cache memory?

→ L1, L2, and L3 are types of cache memory.

Q258. In direct mapping, how is a block of main memory mapped to the cache?

→ Each main memory block maps to exactly one specific cache line.

Q259. What is the main disadvantage of direct mapping?

→ Multiple memory blocks competing for the same cache line causes frequent replacements.

Q260. In set-associative mapping, how is a block mapped to cache?

→ A block maps to a specific set but can go into any line within that set.

Q261. In fully associative mapping, how is a block mapped to cache?

→ A block can be placed in any cache line without restriction.

Q262. What is the purpose of the tag field in cache memory?

→ The tag field identifies which main memory block is stored in a cache line.

Q263. What is the role of the index field in cache memory?

→ The index field identifies which set or line in the cache to access.

Q264. What is the purpose of the cache hit ratio?

→ The hit ratio measures the fraction of accesses found in cache, indicating cache effectiveness.

Q265. Which constraint ensures a column cannot have NULL values?

→ NOT NULL constraint ensures a column cannot have NULL values.

Q266. Which SQL command removes a table from the database?

→ DROP TABLE command removes a table from the database.

Q267. Which is an example of a domain constraint?

→ Specifying that age must be a positive integer is a domain constraint.

Q268. In an ER diagram, what does a double rectangle represent?

→ A double rectangle represents a Weak Entity.

Q269. What is the output of: SELECT Department, COUNT(*) FROM Employee GROUP BY Department?

→ It returns each department name along with the count of employees in that department.

Q270. Which of the following is true about assertions in SQL?

→ Assertions are database-wide constraints that must always be satisfied.

Q271. Which of the following is a type of index?

→ Dense index, sparse index, clustered index, and B-tree index are types of indexes.

Q272. Which normal form is achieved by eliminating transitive dependencies?

→ Third Normal Form (3NF) eliminates transitive dependencies.

Q273. Which SQL data type stores a fixed-length string?

→ CHAR(n) stores a fixed-length string.

Q274. Which is a step in finding the minimal cover of a set of FDs?

→ Eliminate redundant FDs and reduce left-hand sides to single attributes.

Q275. Which of the following is true about a table in 1NF?

→ In 1NF, every attribute contains atomic (indivisible) values.

Q276. Which table is not in 1NF?

→ A table with multi-valued or composite attributes in a single cell is not in 1NF.

Q277. Which is a disadvantage of the Two-Phase Locking (2PL) protocol?

→ 2PL can cause deadlocks.

Q278. Which is a concurrency control protocol?

→ Two-Phase Locking (2PL) and Timestamp Ordering are concurrency control protocols.

Q279. What is the primary goal of concurrency control in a database system?

→ The goal is to ensure isolation and consistency of concurrent transactions.

Q280. Which of the following is a state of a transaction in a database system?

→ Active, Partially Committed, Committed, Failed, and Aborted are transaction states.

Q281. Which technique prevents deadlocks in a database system?

→ Wait-Die and Wound-Wait schemes are techniques to prevent deadlocks.

Q282. Which of the following is a type of lock in a database system?

→ Shared (read) lock and Exclusive (write) lock are types of locks.

Q283. Which is true about the Two-Phase Locking (2PL) protocol?

→ 2PL guarantees conflict serializability.

Q284. What is the main purpose of time complexity analysis?

→ To estimate how the runtime of an algorithm grows as input size increases.

Q285. Which asymptotic notation represents the upper bound?

→ Big-O (O) notation represents the upper bound.

Q286. Sparse matrix is a matrix in which

→ Sparse matrix is one in which most elements are zero.

Q287. Which data structure follows LIFO principle?

→ Stack follows the LIFO (Last In, First Out) principle.

Q288. Which operation is used to insert an element into a stack?

→ The Push operation inserts an element into a stack.

Q289. Which linked list allows traversal in both directions?

→ Doubly Linked List allows traversal in both directions.

Q290. In a circular linked list, the last node points to

→ The last node points to the first node (head).

Q291. Which traversal of a binary tree visits root first?

→ Pre-order traversal visits root first.

Q292. Binary Search Tree property states

→ Left subtree has smaller values and right subtree has larger values than the root.

Q293. Which data structure is best for BFS traversal?

→ A Queue is best for BFS traversal.

Q294. Graphs are commonly used to represent

→ Graphs represent networks, relationships, maps, and social connections.

Q295. Which application uses shortest path algorithms?

→ GPS navigation and network routing use shortest path algorithms.

Q296. Social networks are modeled using

→ Social networks are modeled using graphs.

Q297. DFS uses which underlying principle?

→ DFS uses the Stack (LIFO) principle.

Q298. Which graph traversal is better for finding shortest path in unweighted graph?

→ BFS (Breadth-First Search) is better for shortest path in an unweighted graph.

Q299. Which sorting algorithm has best average case complexity $O(n \log n)$?

→ Merge Sort and Quick Sort have $O(n \log n)$ average case complexity.

Q300. Hashing technique is mainly used for

→ Hashing is mainly used for fast data retrieval and insertion.

Q301. Collision in hashing occurs when

→ Collision occurs when two different keys hash to the same index.

Q302. Which is a collision resolution technique?

→ Chaining and Open Addressing are collision resolution techniques.

Q303. Open addressing is used in

→ Open addressing is used in hash tables to resolve collisions without chaining.

Q304. Which of the following is NOT a function of a DBMS?

→ Compiling programs is NOT a function of a DBMS.

Q305. In three-schema architecture, logical data independence means

→ It means changing the logical schema without affecting external/application schemas.

Q306. Which ER constraint specifies number of entities participating in a relationship?

→ Cardinality constraint specifies the number of participating entities.

Q307. A weak entity set is identified using

→ A weak entity set is identified using a partial key and its owner entity's key.

Q308. Which SQL clause is applied after GROUP BY but before ORDER BY?

→ HAVING clause is applied after GROUP BY and before ORDER BY.

Q309. Hash indexing is inefficient for

→ Hash indexing is inefficient for range queries.

Q310. A relation is in 2NF if it is in 1NF and

→ Every non-prime attribute is fully functionally dependent on the primary key.

Q311. Which index supports ordered traversal of records?

→ B-Tree index supports ordered traversal of records.

Q312. If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$ is called

→ This is called Transitivity (Armstrong's transitivity axiom).

Q313. Which normal form guarantees lossless decomposition based only on FDs?

→ BCNF (Boyce-Codd Normal Form) guarantees lossless decomposition.

Q314. Minimal cover of a set of FDs has

→ Minimal cover has no redundant FDs and no extraneous attributes.

Q315. A relation in BCNF may still have

→ A relation in BCNF may still have multi-valued dependencies (4NF issues).

Q316. Which property ensures database consistency after transaction completion?

→ Durability (of ACID properties) ensures consistency after transaction completion.

Q317. A schedule is conflict serializable if

→ It can be transformed into a serial schedule by swapping non-conflicting operations.

Q318. Dirty read problem occurs when

→ A transaction reads data written by another uncommitted transaction.

Q319. Which recovery method does NOT require logging?

→ Shadow Paging does not require logging.

Q320. Two-phase locking protocol has which phases?

→ Growing phase (acquiring locks) and Shrinking phase (releasing locks).

Q321. Strict 2PL avoids

→ Strict 2PL avoids cascading aborts (dirty reads).

Q322. Which protocol guarantees conflict serializability?

→ Two-Phase Locking (2PL) guarantees conflict serializability.

Q323. Checkpointing mainly reduces

→ Checkpointing mainly reduces recovery time after a crash.

Q324. $L = \{a^nk \mid k > 0, n \text{ is a positive integer constant}\}$. Minimum number of states in DFA?

→ $n+1$ states are needed in the DFA.

Q325. Which language over $\{0,1\}$ is described by $(0+1)^*0(0+1)^*0(0+1)^*$?

→ All binary strings containing at least two 0s.

Q326. Which page replacement algorithm suffers from Belady's Anomaly?

→ FIFO (First-In-First-Out) page replacement suffers from Belady's Anomaly.

Q327. FIFO page replacement, 3 frames: A,B,C,D,A,B,E,A,B,C,D,E — number of page transfers?

→ 9 page faults occur with 3 frames using FIFO.

Q328. FIFO, 4 frames: A,B,C,D,A,B,E,A,B,C,D,E — number of page transfers?

→ 10 page faults occur with 4 frames (demonstrating Belady's Anomaly).

Q329. P: Increasing frames sometimes increases page fault rate. Q: Some programs have no locality. TRUE?

→ Both P and Q are TRUE.

Q330. Which statements are TRUE for Shared Memory and Message Passing IPC?

→ Statements I, II, and V are TRUE.

Q331. Which of the following are TRUE for direct communication?

→ Processes must name each other explicitly, and exactly one link exists between each pair.

Q332. In indirect communication between processes P and Q

→ P and Q communicate via a shared mailbox (port); multiple processes can share it.

Q333. In the Zero capacity queue

→ The sender must block until the receiver is ready (no buffering; rendezvous required).

Q334. Correct statements regarding mutex lock to prevent race condition.

→ Statements 1 and 3 are correct: acquire lock before entering, release when exiting.

Q335. Indefinite blocking may occur if we remove processes from semaphore list in _____ order.

→ Indefinite blocking may occur with non-FIFO (e.g., LIFO) order.

Q336. In Priority Inversion, which stands TRUE?

→ A lower-priority task holding a resource needed by a higher-priority task causes priority inversion.

Q337. A set of waiting processes where P_0 waits for P_1 , ..., P_n waits for P_0 . This deadlock condition is?

→ This condition is called Circular Wait.

Q338. Which statements are TRUE for dealing with deadlock?

→ All three statements are TRUE — prevention, detection/recovery, and ignoring are all valid approaches.

Q339. System has 12 tape drives, P_0 needs 10, P_1 needs 4, P_2 needs 9. Allocated: 5,2,2. Safe sequence?

→ Available = 3; P_1 can finish (needs 2 more), releasing 4; then P_0 (needs 5, gets 7); then P_2 . Safe sequence: P_1, P_0, P_2 .

Q340. 46-bit virtual, 32-bit physical, 3-level page table, PTE=32 bits, 1MB 16-way set-associative cache, 64-byte block. Page size?

→ Page size is 4 KB.

Q341. FIFO, 4 frames, access 100 distinct pages, then 100 distinct pages, then 100 in reverse. Page faults?

→ 200 page faults occur.

Q342. LRU, FIFO, Optimal for reference string 1,2,3,2,4,1,3,2,4,1 with 3 frames.

→ LRU = 6 faults, FIFO = 7 faults, Optimal = 5 faults.

Q343. Which of the following needs a device driver?

→ Disk drives, printers, and other hardware devices need device drivers.

Q344. The circuit used to store one bit of data is known as

→ A Flip-Flop is the circuit used to store one bit of data.

Q345. Rotational latency refers to

→ The time for the desired disk sector to rotate under the read/write head.

Q346. (2FAOC)16 is equivalent to

→ $(2FAOC)_{16} = 0010\ 1111\ 1010\ 0000\ 1100$ in binary (196,108 in decimal).

Q347. Which of the following is not a weighted code?

→ Excess-3 code is not a weighted code.

Q348. A Stack-organised Computer uses instruction of

→ A stack-organised computer uses zero-address instructions.

Q349. Main memory 8K bytes, cache 2K words, associative mapping. Each word of cache shall be

→ Each cache word stores the data plus the full main memory address tag.

Q350. An n-bit microprocessor has

→ An n-bit microprocessor has n-bit data bus, n-bit ALU, and n-bit registers.

Q351. When CPU executes a program that is part of the OS, it is said to be in

→ The CPU is said to be in Kernel Mode (Supervisor Mode).

Q352. What is the value of n in multiplication of $110 * 1000$?

→ $110 * 1000 = 110000$; $n = 6$ (result has 6 bits, with 3 zeros appended).

Q353. If the binary point is fixed to the right of the rightmost digit, the representation is called

→ It is called Integer Representation (Fixed-point with no fractional part).

Q354. In Reverse Polish notation, $A*B+C*D$ is written as

→ $A B * C D * +$ is the Reverse Polish (postfix) notation.

Q355. Multiplicand $B=10010$, Multiplier $Q=10110$. How many add operations in multiplication?

→ 3 add operations are performed (for the three 1-bits in multiplier 10110).

Q356. Final product of multiplication stored in

→ The final product is stored in the combined A (AC) and Q (QR) register pair.

Q357. If QR (Multiplier) is 10-bit signed 2's complement, value of SC at initial phase of Booth multiplication?

→ SC (step counter) is initialized to 9 ($n-1$ for $n=10$ bits).

Q358. For what value of $Q_n Q_{n+1}$ is addition of AC & BR performed in Booth multiplication?

→ Addition is performed when $Q_n Q_{n+1} = 10$.

Q359. Which facilitates transfer of bulk data from hard disk to main memory with highest throughput?

→ DMA (Direct Memory Access) provides the highest throughput for bulk data transfer.

Q360. Which will be handled at the highest priority?

→ Non-maskable hardware interrupts (NMI) are handled at the highest priority.

Q361. What does the hardwired control generator consist of?

→ It consists of an instruction decoder, sequencer, and combinational logic circuits.

Q362. Devices A (ID=5) and B (ID=6) contesting for BUS. Which gets the bus via distributed arbitration?

→ Device B (ID=6) gets the bus since it has the higher ID.

Q363. Cache Memory is implemented using DRAM chips. True or False?

→ False — Cache memory is implemented using SRAM (Static RAM) chips, not DRAM.

Q364. Main memory has 2^m blocks, cache has 2^c blocks, set associative with 2 blocks/set. Block k maps to set?

→ Block k maps to set $(k \bmod 2^{(c-1)})$.

Q365. In k-way set associative cache, main memory block j maps to cache lines from?

→ Block j maps to lines $(j \bmod v)*k$ to $(j \bmod v)*k + (k-1)$.

Q366. Given basic ER and relational models, which is INCORRECT?

→ C) In a row of a relational table, an attribute can have more than one value — this is INCORRECT.

Q367. Given basic ER and relational models, which is CORRECT?

→ B) An attribute of an entity can be composite is CORRECT.

Q368. Which of the following is not an example of DBMS?

→ Google Search is NOT an example of a DBMS.

Q369. Which is not a function of the database?

→ D) Analysing code is not a function of a database.

Q370. Which statements are TRUE about an SQL query with HAVING/GROUP BY?

→ P and S are TRUE: HAVING can exist without GROUP BY; not all GROUP BY attributes need to be in SELECT.

Q371. `SELECT studname WHERE marks > SOME (SELECT marks FROM student WHERE SECTION='c')`

→ B) Gives all studnames where marks are greater than at least one student in section c.

Q372. `WITH max_marks AS (SELECT MAX(marks) FROM student) SELECT studentID WHERE marks = max_marks.value`

→ B) The query gives the studentID of the student with the maximum marks.

Q373. employee decomposed into employee1(ID,name) and employee2(name,street,city,salary). Type?

→ This is a Lossy (lossy-join) decomposition.

Q374. Inst_dept decomposed into instructor and department. Type?

→ This is a Lossless-join decomposition.

Q375. $R(A,B,C,D,E)$ with $ABC \rightarrow DE$ and $D \rightarrow AB$. Number of superkeys?

→ The number of superkeys of R is 8.

Q376. Which of the two SQL queries I1 and I2 correctly finds employees managed by those managed by emp 123?

→ Both I1 and I2 are correct and equivalent in SQL.

Q377. Which of the four schedules is conflict serializable?

→ The schedule that can be transformed to a serial schedule by swapping non-conflicting operations is conflict serializable.

Q378. Consider schedule S of T1,T2,T3,T4.

→ A schedule is conflict serializable if its conflict graph (precedence graph) is acyclic.

Q379. Which statement about schedules S1 and S2 is TRUE?

→ S1 is conflict serializable and S2 is not conflict serializable.

Q380. Two-phase locking protocol with exclusive locks in increasing address order. This protocol will

→ This protocol will prevent deadlocks and ensure conflict serializability.

Q381. R-timestamp(Q) denotes?

→ R-timestamp(Q) denotes the timestamp of the last transaction that read data item Q.

Q382. Number of different locks used in multi-version two-phase locking is equal to

→ It is equal to 2 (shared and exclusive locks).

Q383. Which locking protocol obtains all locks at beginning and releases all at end?

→ Conservative Two-Phase Locking (static locking) obtains all locks at start and releases all at end.

Q384. Checkpoints are a part of

→ Checkpoints are a part of the database recovery mechanism.

Q385. Space complexity of a recursive algorithm with depth 'd'?

→ Space complexity is $O(d)$ due to the recursive call stack.

Q386. Time complexity of inserting at end of singly linked list?

→ $O(n)$ if no tail pointer; $O(1)$ if a tail pointer is maintained.

Q387. Time complexity of the linear search algorithm?

→ $O(n)$ is the time complexity of linear search.

Q388. Time complexity of finding an element in a binary search tree?

→ $O(h)$ where h is the height; $O(\log n)$ on average, $O(n)$ worst case.

Q389. Time complexity of inserting a node at beginning of a singly linked list?

→ $O(1)$ — inserting at the beginning of a singly linked list takes constant time.

Q390. In a doubly linked list, how many pointers does each node contain?

→ Each node contains 2 pointers: one to the next node and one to the previous node.

Q391. In binary tree array representation, left child of node at index i is at?

→ Left child is at index $2i+1$ (0-indexed) or $2i$ (1-indexed).

Q392. Time complexity of BFS on graph with V vertices and E edges?

→ $O(V + E)$ is the time complexity of BFS.

Q393. Which graph traversal guarantees visiting every node exactly once?

→ Both BFS and DFS guarantee visiting every node exactly once.

Q394. Which algorithm finds strongly connected components?

→ Tarjan's or Kosaraju's algorithm finds strongly connected components.

Q395. Which data structure is NOT typically used to represent a graph?

→ A heap is NOT typically used to represent a graph.

Q396. Which algorithm finds shortest path with negative edge weights?

→ Bellman-Ford algorithm finds shortest path with negative edge weights.

Q397. Which sorting algorithm has worst-case $O(n^2)$?

→ Bubble Sort, Insertion Sort, and Selection Sort have $O(n^2)$ worst case.

Q398. Which sorting algorithm always requires more than $O(n \log n)$ time?

→ No comparison-based sort can do better than $O(n \log n)$; none always requires more.

Q399. Which is NOT a characteristic of a good hash function (2)?

→ A good hash function should NOT be slow to compute.

Q400. Which collision resolution technique uses double hashing?

→ Double Hashing resolves collisions by rehashing with a different hash function.

Q401. Which sorting algorithm has worst-case $O(n^2)$ but preferred for small datasets?

→ Insertion Sort has $O(n^2)$ worst case but is preferred for small datasets.

Q402. Which collision resolution allows load factor to exceed 1?

→ Separate Chaining allows load factor to exceed 1 without performance degradation.

Q403. In which sorting technique is best-case also $O(n \log n)$?

→ Merge Sort has $O(n \log n)$ in all cases (best, average, and worst).

Q404. In linear probing, when collision occurs, how is next slot searched?

→ The next slot is found by checking $(hash + 1) \bmod m$, $(hash + 2) \bmod m$, etc. (sequential search).

Q405. Formal definition of a DFA?

→ DFA is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where $\delta: Q \times \Sigma \rightarrow Q$ is deterministic.

Q406. Primary difference between DFA and NFA?

→ DFA has exactly one transition per input; NFA can have zero or more transitions.

Q407. In DFA, if $\delta(q, a) = p$, it implies that?

→ From state q on input a , the machine deterministically moves to state p .

Q408. Which is true regarding NFA with ϵ transitions?

→ ϵ -transitions allow state changes without consuming any input symbol.

Q409. In NFA with ϵ transitions, if $\delta(q, \epsilon)$ contains multiple states, what does it indicate?

→ The machine can be in any of those states simultaneously without reading input.

Q410. In regular grammars, production rules are of the form?

→ Rules are of the form $A \rightarrow aB$ or $A \rightarrow a$ (right-linear or left-linear).

Q411. Regular languages are closed under which operations?

→ Regular languages are closed under union, intersection, complement, concatenation, and Kleene star.

Q412. Pumping Lemma states for regular languages that any string longer than p can be?

→ Any string longer than p can be pumped (the middle portion repeated any number of times).

Q413. CNF requires that?

→ CNF requires every production to be of the form $A \rightarrow BC$ or $A \rightarrow a$.

Q414. Ambiguity in CFGs?

→ Ambiguity means a string has more than one leftmost derivation (multiple parse trees).

Q415. A derivation tree?

→ A derivation tree (parse tree) represents the hierarchical structure of a derived string.

Q416. Myhill-Nerode Theorem is used to?

→ It is used to determine if a language is regular and to minimize DFA states.

Q417. Transition function of NPDA is of the form?

→ $\delta: Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \rightarrow 2^*(Q \times \Gamma^*)$.

Q418. Conversion of CFG to PDA involves?

→ It involves creating a PDA that simulates leftmost derivations of the grammar.

Q419. Pumping Lemma for CFLs: if L is CFL there exists p such that?

→ Any string s in L with $|s| \geq p$ can be written as $s = uvxyz$ with $|vxy| \leq p$, $|vy| \geq 1$, and $uv^i xy^i z \in L$ for all $i \geq 0$.

Q420. Production rules in CSG are of the form?

→ Rules are of the form $\alpha A \beta \rightarrow \alpha \gamma \beta$ where A is a non-terminal and $|\gamma| \geq 1$.

Q421. Universal Turing Machine is capable of?

→ A Universal Turing Machine can simulate any other Turing Machine.

Q422. Halting Problem for Turing Machines refers to?

→ The problem of determining whether a given TM halts on a given input — it is undecidable.

Q423. A language is recursively enumerable if (2)?

→ A TM accepts it; the TM may loop forever on inputs not in the language.

Q424. In Spinlocks?

→ Spinlocks cause busy-waiting: a process repeatedly checks the lock without sleeping.

Q425. $T(n) = 2T(n/2) + O(n)$. Time complexity using Master Theorem?

→ $O(n \log n)$ by Master Theorem Case 2.

Q426. Which complexity grows fastest as n increases?

→ Factorial $O(n!)$ grows the fastest.

Q427. How many additions to add two $n \times n$ sparse matrices in linked list representation?

→ $O(n^2)$ additions in the worst case (proportional to non-zero elements).

Q428. Evaluate postfix: $5\ 3\ 4\ * + 9\ -$

→ $5 + (3 * 4) - 9 = 5 + 12 - 9 = 8$.

Q429. Sparse matrix $n \times n$ with at most k non-zero elements. Optimal space complexity?

→ $O(k)$ space complexity is optimal for storing k non-zero elements.

Q430. $T(n) = 3T(n/4) + O(n^2)$. Asymptotic complexity?

→ $O(n^2)$ by Master Theorem Case 3 (since n^2 dominates).

Q431. Circular queue of size n. Condition to check if queue is full?

→ $(\text{rear} + 1) \bmod n == \text{front}$ is the condition for a full circular queue.

Q432. Worst-case time complexity of inserting into a sorted doubly linked list of n elements?

→ $O(n)$ — must traverse to find the correct position.

Q433. Space complexity of adjacency matrix for undirected graph with V vertices and E edges?

→ $O(V^2)$ space complexity for adjacency matrix representation.

Q434. Complete binary tree of height h contains how many nodes?

→ A complete binary tree of height h has between 2^h and $2^{h+1} - 1$ nodes.

Q435. Number of minimum spanning trees in a connected weighted graph with distinct edge weights?

→ Exactly 1 — distinct edge weights guarantee a unique MST.

Q436. Connected graph with V vertices and V-1 edges. Maximum number of spanning trees?

→ Exactly 1 spanning tree (it is already a tree).

Q437. Which algorithm detects a negative weight cycle?

→ Bellman-Ford algorithm detects negative weight cycles.

Q438. If MST has weight W and a new edge of weight X is added, what happens?

→ If X is less than the maximum edge in the cycle formed, the MST may change; otherwise MST remains the same.

Q439. A graph is bipartite if and only if?

→ A graph is bipartite if and only if it has no odd-length cycles.

Q440. If n keys distributed into m slots uniformly, expected keys per slot?

→ Expected keys per slot = n/m (the load factor λ).

Q441. A perfect hash function ensures?

→ A perfect hash function ensures no collisions — each key maps to a unique slot.

Q442. Which sorting algorithm runs in $O(n)$ for already sorted input?

→ Insertion Sort runs in $O(n)$ for an already sorted array.

Q443. Which sorting algorithm has best worst-case performance?

→ Merge Sort has the best guaranteed worst-case performance of $O(n \log n)$.

Q444. Hash table with chaining, load factor λ . Average search time (worst case)?

→ $O(1 + \lambda)$ average time complexity for search with chaining.

Q445. Worst-case time to insert n elements into empty binary heap?

→ $O(n \log n)$ for n insertions; $O(n)$ if using build-heap (heapify).

Q446. 300 processes/sec, 5 system calls each, 2ms per call. Total system call execution time/sec?

→ $300 \times 5 \times 2\text{ms} = 3000\text{ms} = 3$ seconds total system call time per second.

Q447. 4 processes in round-robin, 5ms quantum, 10ms CPU + 5ms I/O cycles. Context switches in one cycle?

→ Each process is scheduled multiple times; approximately 8 context switches occur per cycle.

Q448. `fork()` called, then `if(fork()) printf`. How many times is 'Hello' printed?

→ 'Hello' is printed 2 times.

Q449. Microkernel moves 80% services to user mode, 10% overhead per call, 1000 system calls?

→ Additional overhead = $1000 \times 10\% = 100$ extra service call overheads.

Q450. Layered OS with 5 layers, 2ms delay per layer, original 3ms. Total time per system call?

→ $3\text{ms} + 5 \times 2\text{ms} = 13\text{ms}$ total.

Q451. Time slice 50ms, context switch 5ms, 5 processes. Fraction of CPU wasted?

→ $5\text{ms} / (50\text{ms} + 5\text{ms}) \approx 9.09\%$ of CPU time wasted per process switch.

Q452. 5 processes burst times {10,5,8,3,2}ms. FCFS average waiting time?

→ Average waiting time = $(0+10+15+23+26)/5 = 74/5 = 14.8\text{ms}$.

Q453. Shared variable `count++` and `count--` run concurrently. What issue arises?

→ A race condition arises, causing incorrect values of count.

Q454. Round-robin, 4ms quantum, process needs 20ms. How many times scheduled?

→ The process will be scheduled 5 times.

Q455. A mutex lock implementation ensures?

→ It ensures mutual exclusion so only one thread executes the critical section at a time.

Q456. Semaphore `mutex=1`; `wait(mutex)`; `critical_section()`; `signal(mutex)`. Two processes simultaneously?

→ One process enters the critical section; the other blocks until the first exits.

Q457. A binary semaphore can take values?

→ A binary semaphore can take values 0 or 1.

Q458. Producer-consumer, buffer size 5, 10 items produced and consumed sequentially. Full buffer occurrences?

→ The buffer becomes full 2 times (after every 5 items produced without consumption).

Q459. Monitors for synchronization ensures?

→ Monitors ensure mutual exclusion and structured condition synchronization automatically.

Q460. A semaphore value of -3 indicates?

→ A semaphore value of -3 indicates 3 processes are waiting for the resource.

Q461. 4 processes, max claims {10,4,9,7}, available=7. Banker's Algorithm safe sequence?

→ Yes, a safe sequence exists (e.g., $P2 \rightarrow P4 \rightarrow P1 \rightarrow P3$).

Q462. 4KB page size, 64MB physical memory. How many frames?

→ $64\text{MB} / 4\text{KB} = 16,384$ frames available.

Q463. 12 page faults for 3-frame, 9 faults for 4-frame. Which anomaly?

→ This is Belady's Anomaly (fewer faults with more frames — opposite of expected with FIFO).

Q464. Process requires 100KB but allocated 120KB. Type of fragmentation?

→ Internal fragmentation occurs (allocated space exceeds actual need).

Q465. Paging with 3-level page table, 10-bit entries per level. Max virtual address space?

→ Max virtual address space = $2^{(10+10+10+12)} = 2^{42}$ bytes = 4TB (assuming 12-bit page offset).

Q466. Which represents language with no pair of consecutive 1's if $\Sigma=\{0,1\}$?

→ The regular expression $(0+10)^*(1+\epsilon)$ represents strings with no consecutive 1s.

Q467. Total memory locations addressable by 16-bit address bus?

→ $2^{16} = 65,536$ memory locations (64KB).

Q468. 32-bit address bus, 16-bit data bus. Maximum addressable memory?

→ $2^{32} = 4\text{GB}$ maximum addressable memory.

Q469. 24-bit address bus. Maximum addressable memory in bytes?

→ $2^{24} = 16,777,216$ bytes = 16MB.

Q470. Which is an example of inter-register transfer?

→ $R1 \leftarrow R2$ (moving data from one register to another) is an inter-register transfer.

Q471. ALU: A=1010, B=0110, operation=subtraction. Output?

→ $1010 - 0110 = 0100$ (decimal $10 - 6 = 4$).

Q472. 4-stage pipeline, 2ns per stage. Time to execute 100 instructions?

→ Time = $(4 + 99) \times 2\text{ns} = 206\text{ns}$.

Q473. 4-stage pipeline, 10 instructions. Minimum cycles required?

→ Minimum cycles = $4 + (10-1) = 13$ cycles.

Q474. Direct-mapped cache: 32KB size, 64B block, 1GB main memory. Tag, index, offset bits?

→ Offset=6 bits, Index=9 bits, Tag=15 bits (32-bit address: $32-6-9=17$; check: $\log_2(32\text{KB}/64\text{B})=9$).

Q475. LRU cache, 4 blocks, accesses: 1,2,3,4,1,2,5,1,2,3,4,5. Cache misses?

→ 8 cache misses occur.

Q476. FIFO page replacement, 3 frames, pages: 1,2,3,4,1,2,5,1,2,3,4,5. Page faults?

→ 9 page faults occur.

Q477. SELECT COUNT(*) FROM EMPLOYEE WHERE Department='Sales'?

→ Returns the number of employees in the Sales department.

Q478. SELECT Name FROM Employee WHERE Salary > (SELECT AVG(Salary) FROM Employee)?

→ Returns names of employees whose salary is above the average salary.

Q479. Transaction T is in partially committed state. Which is true?

→ T has executed its last operation but the changes are not yet permanently stored to disk.

Q480. Wait-for graph: T1→T2→T3→T1. Is there deadlock?

→ Yes, there is a deadlock — the circular wait indicates a deadlock.

Q481. After crash, log entries present. Which transactions need redo?

→ Transactions that have a commit record in the log but changes not yet on disk need to be redone.

Q482. Logical data independence refers to?

→ The ability to change the logical schema without affecting application programs or external views.

Q483. Which DBMS component manages data storage on disk?

→ The Storage Manager (or File Manager) manages data storage on disk.

Q484. A weak entity set always has?

→ A weak entity set always has a partial key and depends on an owner (strong) entity set.

Q485. Total participation in ER model means?

→ Every entity in the entity set must participate in at least one relationship instance.

Q486. Which SQL clause filters groups after aggregation?

→ HAVING clause filters groups after aggregation.

Q487. Which index is most suitable for range queries?

→ B-Tree index is most suitable for range queries.

Q488. A relation in 1NF must have?

→ All attributes must have atomic (indivisible) values with no repeating groups.

Q489. Which SQL object enforces automatic actions on data changes?

→ A Trigger enforces automatic actions on data changes.

Q490. If $X \rightarrow Y$ holds, then Y is?

→ Y is functionally dependent on X.

Q491. Which normal form removes transitive dependency?

→ Third Normal Form (3NF) removes transitive dependencies.

Q492. BCNF requires that for every FD $X \rightarrow Y$, X must be?

→ X must be a superkey.

Q493. Minimal cover of FDs is used mainly for?

→ Minimal cover is used mainly for normalization and schema design.

Q494. Which ACID property ensures all-or-nothing execution?

→ Atomicity ensures all-or-nothing execution.

Q495. A schedule is conflict serializable if (2)?

→ Its precedence (conflict) graph is acyclic.

Q496. Dirty read occurs when a transaction reads?

→ Dirty read occurs when a transaction reads uncommitted data from another transaction.

Q497. Which recovery method uses before-image and after-image logs?

→ Deferred update and immediate update recovery methods use before and after images.

Q498. Two-phase locking has which two phases?

→ Growing phase (lock acquisition) and Shrinking phase (lock release).

Q499. Strict 2PL prevents?

→ Strict 2PL prevents cascading rollbacks (dirty reads).

Q500. Timestamp ordering protocol avoids?

→ Timestamp ordering avoids deadlocks by assigning transaction timestamps.

Q501. Checkpointing primarily reduces (2)?

→ Checkpointing primarily reduces the amount of log to be scanned during recovery.

Q502. Which SQL query retrieves all rows from STUDENT?

→ SELECT * FROM STUDENT;

Q503. Which clause filters rows before grouping?

→ WHERE clause filters rows before grouping.

Q504. Which query counts rows in EMP table?

→ SELECT COUNT(*) FROM EMP;

Q505. Which SQL keyword removes duplicate rows?

→ DISTINCT keyword removes duplicate rows.

Q506. Which query finds average salary department-wise?

→ SELECT Department, AVG(Salary) FROM EMP GROUP BY Department;

Q507. HAVING clause is used to filter?

→ HAVING is used to filter groups after aggregation.

Q508. Which query finds departments with more than 5 employees?

→ SELECT Department FROM EMP GROUP BY Department HAVING COUNT(*) > 5;

Q509. GROUP BY is mandatory when using?

→ GROUP BY is mandatory when using aggregate functions with grouping (with HAVING).

Q510. A subquery is a query written?

→ A subquery is written inside another query (nested within SELECT, FROM, or WHERE clause).

Q511. Which operator is used with subquery returning multiple values?

→ IN operator is used with a subquery returning multiple values.

Q512. Which subquery executes once for entire query?

→ A non-correlated (simple) subquery executes once for the entire outer query.

Q513. Correlated subquery is evaluated?

→ A correlated subquery is evaluated once for each row of the outer query.

Q514. Which join returns only matching rows?

→ INNER JOIN returns only matching rows from both tables.

Q515. Which join returns all rows from left table?

→ LEFT OUTER JOIN returns all rows from the left table.

Q516. Natural join matches tables based on?

→ Natural join matches tables based on columns with the same name and data type.

Q517. Which join produces Cartesian product?

→ CROSS JOIN produces the Cartesian product of two tables.

Q518. Which query returns highest salary from EMP?

→ SELECT MAX(Salary) FROM EMP;

Q519. EXISTS operator checks?

→ EXISTS checks whether a subquery returns any rows (returns TRUE if non-empty).

Q520. ORDER BY sorts result by default in?

→ ORDER BY sorts results in ascending (ASC) order by default.

Q521. Which keyword limits number of rows in SQL?

→ LIMIT (or TOP in SQL Server, ROWNUM in Oracle) limits the number of rows returned.